

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

3rd December 2020

Subject SP4 – Pensions and Other Benefits

Time allowed: 3 Hours 30 Minutes (09.30 – 13.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
- 3. Mark allocations are shown in brackets.*

- Q. 1)** A company is planning to set up a Defined Benefit Pension Scheme in order to improve recruitment and retention of employees. The Company has approached you to provide guidance in designing the scheme to meet the objective.
- i) Describe the main design elements the company should consider when designing the scheme. (7)
- One of the suggestions by you is the minimum period of employment to join the scheme. The Finance Director wants to understand it further.
- ii) Discuss the advantages and disadvantages in having a waiting period before employees become eligible to join their retirement benefit scheme. (4)
- The Finance Director is concerned about the risk of future costs arising as a result of this scheme.
- iii) Outline the main ways in which the scheme could be set up that would reduce these risks. (8)
- Having gone through your note to reduce the cost of the scheme, he finds a scheme that provides lumpsum at retirement to the members.
- iv) Explain the features of such scheme. (3)
- Employees are very keen to have flexibility around the benefits they receive from the scheme.
- v) List four benefit options that could be provided to members by the scheme. (2)
- vi) Outline the risks that the company might face as a result of introducing these member options. (4)
- vii) Outline the approaches a scheme's trustees can use in order to monitor the strength of the sponsor covenant and describe the actions the trustees are likely to be able to take if the sponsor is "in distress". (5)
- [33]**
- Q. 2)** The Finance Director of a newly established defined benefit scheme has asked you to prepare a note to cover financing and investment strategy of the scheme.
- i) Define the criteria used to choose the funding strategy for funding future benefit payments. (3)
- ii) Outline the considerations for determining an appropriate investment strategy if the scheme is to be funded. (9)
- The debt market of the Country has been very strong and providing good returns. The Finance Director asks you if the funds can be invested in fixed interest securities. The Trustee, however, raises concern on this as the liabilities are real in nature.
- iii) Give reasons why a defined benefit pension scheme may hold domestic fixed-interest securities. (5)
- [17]**
- Q. 3)** You are the actuary for a large private sector bank in India, which sponsors a contributory final salary (DB) pension scheme. For the last few years, the actuarial valuation of the scheme has consistently revealed a significant deficit. The scheme retains deferred and pensioners liabilities within the fund and does not buy out annuities from insurance companies.

- i) Describe briefly, and comment on the suitability of, the possible courses of action that a scheme may take when a deficit is revealed following the valuation of a final salary (DB) pension scheme. (8)

This year, your client informs you that they have decided to close the contributory final salary (DB) pension scheme to future accrual and replace it with a Defined Contribution (DC) superannuation scheme. You have been asked to advise the bank on the closure of the DB scheme and the design of the new DC scheme.

- ii) The Head of Treasury has asked you to explain the impact of the closure on the investment policy followed in the DB scheme. Outline the points you would make in your reply. (9)
- iii) The HR Head is considering whether to let members contribute to the DC scheme, as the DB scheme was contributory. Discuss the factors to be considered in deciding whether or not members should be required to contribute to the DC scheme, and if so the level of member contributions required. (8)

The employer has asked you to carry out projections of the likely level of normal retirement benefits from the DC scheme for a group of employees within the bank.

- iv) List the assumptions that need to be made in carrying out the projections. (3)

One of the Trustees has proposed that in order to provide more flexibility, members should have a choice of funds within the DC scheme.

- v) Describe the considerations that should be taken into account in determining the investments to be offered. (8)
- [36]

- Q. 4)** You are a retirement consultant in India, where employer-sponsored pension schemes are set up under trust law. Trustees are responsible for securing members' benefits, but it is unclear who is responsible for setting the investment strategy. You are also responsible for conducting training sessions for trustees on a regular basis.

The trustees of a large pension scheme which has a Defined Benefit structure with a Defined Contribution underpin are concerned about the investment strategy. The sponsoring employer determined the current strategy with the scheme assets highly concentrated towards equities.

- i) Explain why the trustees and employer might disagree about the investment strategy and how an asset liability modelling (ALM) exercise can help the trustees to identify a suitable investment strategy. (10)
- ii) Outline the additional assumptions required to calculate asset and liability values as part of an asset liability modelling exercise compared with the assumptions required for a funding valuation. (4)
- [14]
